



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
B.TECH - Semester - VI (Computer Engineering)
SUBJECT: (CE 614) THEORY OF AUTOMATA & FORMAL LANGUAGES

Examination : 3rd Sessional
 Date : 25/03/2023
 Time : 1. Hr. 15 Min.

Seat No. :
 Day :
 Max. Marks :

CE137
 Saturday
 36

INSTRUCTIONS:

- Figures to the right indicate maximum marks for that question.
- The symbols used carry their usual meanings.
- Assume suitable data, if required & mention them clearly.
- Draw neat sketches wherever necessary.

Q.1 Do as Directed

- | | | |
|-------|---|---|
| CO4-R | a. State Church-Turing Thesis and its importance. | 2 |
| CO4-A | b. Using a block diagram, specify the input and output of Universal Turing Machines. | 2 |
| CO4-N | c. We don't talk about $\wedge$ -transition in a Turing Machine. Why are they significant in NFA- $\wedge$ s or in PDAs but not necessary in Turing Machines? | 2 |
| CO3-R | d. Define the rules for Deterministic Push-Down Automata. | 3 |
| CO1-A | e. What is the Left-Recursion problem? Remove Left-Recursion from the given grammar.
$E \rightarrow E + T \mid T$
$T \rightarrow T * F \mid F$
$F \rightarrow a$ | 3 |

Q.2 Answer the Following (Any Two)

- | | | |
|-------|--|-----------|
| CO3-N | a. Construct a Turing Machine for accepting the below given language.
$L = \{ss \mid s \in \{a,b\}^*\}$ | [12]
6 |
| CO3-A | b. Construct a Turing Machine for computing a characteristic function for the language of non-palindromes. | 6 |
| CO3-U | c. With a suitable example, explain cascading of Turing Machines. | 6 |

Q.3 Answer the Following

- | | | |
|-------|--|-----------|
| CO4-N | a. Find an NPDA on $\Sigma = \{a, b, c\}$ that accepts the language given below.
$L = \{w_1cw_2 \mid w_1, w_2 \in \{a, b\}^*, w_1 \neq w_2^R\}$ | [12]
6 |
| CO4-A | b. Construct Top-Down NPDA for the given Grammar
$S \rightarrow aABB \mid aAA, A \rightarrow aBB \mid a, B \rightarrow bBB \mid A$ | 6 |

OR

Q.3 Answer the Following

- | | | |
|-------|---|-----------|
| CO4-N | a. Develop NPDA for the Regular Expression: abb^*aba^* | [12]
6 |
| CO4-A | b. Create Bottom-Up NPDA for the given Grammar
$S \rightarrow aABB \mid aAA, A \rightarrow aBB \mid a, B \rightarrow bBB \mid A$ | 6 |

